Bio-degradable polymers – Drug Delivery

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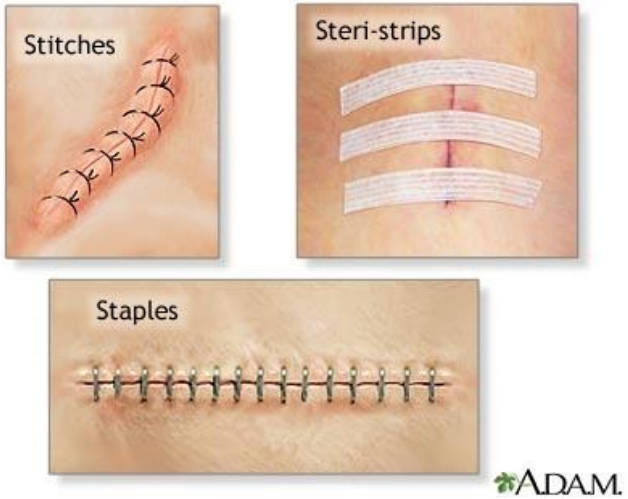
Polyethylene Glycol 400 &
Propylene Glycol Eye Drops
PEGTEARS™
Lubricant Eye Drops
पेगटीयर्स
FOR DRYNESS
AND IRRITATION
10 ML
Alembic
Eye Care

Bio-degradable polymers for Sustainable drug release

- Sustained release allows delivery of a specific drug at a programmed rate that leads to drug delivery for a prolonged period of time.
- This approach of drug release is especially useful for drugs that are metabolized too fast and are eliminated from the body shortly after administration.
- Sustained release by adjusting the speed of drug release can keep the concentration of the drug at a constant level in the blood or target tissue.

Bio-degradable Sutures



Degradable PGA suture material. This particular product is called Dexon

Suture Type	Product Description	Breaking Strength Retention (BSR) Profile	Absorption Rate
VISORB QUICK®	Fast Absorbabing, Coated, Braided, Polyglycolic Acid (PGA), Synthetic Absorbable	54% @ 7 days 50% @ 9-11 days	42 days
VISORB®	Coated, Braided, Polyglycolic Acid (PGA), Synthetic Absorbable	83% @ 14 days 52% @ 21 days 20% @ 28 days	60-90 days
MONOSWIFT®	Monofilament, Poly(glycolide-co-caprolactone) 25 (PGCL), Synthetic Absorbable	70% @ 7 days 40% @ 14 days 12% @ 21 days	91-120 days
MONO-DOX®	Monofilament, Polydioxanone (PDO), Synthetic Absorbable	75% @ 14 days 66% @ 28 days 53% @ 42 days	180 days
CHROMIC GUT	Twisted Gut, Natural Absorbable	Variable*	Variable* (generally 14-28 days)
PLAIN GUT	Twisted Gut, Natural Absorbable	Variable*	Variable* (generally 5-7 days)

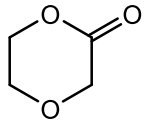
*Refer to package insert

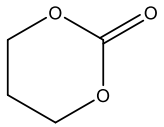
https://openwetware.org/wiki/Degradable_Sutures,_by_Cody_Siroka

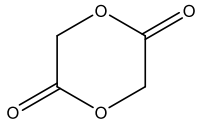
Bio-degradable Sutures

Poly Glycolic Acid (PGA)

Bio-degradable Sutures


dioxanone


trimethylene carbonate


glycolide

•COVIDIEN Biosyn synthetic absorbable sutures are prepared from Gylcomer 631, a synthetic polyester composed of glycolide (60%), dioxanone (14%), and trimethylene carbonate (26%). These sutures are colored violet to increase visibility and are also available undyed.

The advanced extrusion process of the molecule of Gylcomer 631 gives the suture:
•Excellent initial strength
•Excellent in-vivo strength over the critical wound healing period
•Excellent knot security
•Minimal memory for handling ease

Biosyn sutures are indicated for use in general soft tissue approximation and/or ligation including use in ophthalmic surgery, but not for use in cardiovascular or neurological surgery.

Breathing Polymers

Fresh fruits and vegetables are of greater interest also offering good healthy option.
But most of the packing materials used are non bio-degredable which increases the environmental foot print at the same time increases the cost.

Recently Spain based research group developed new polymeric material PLA4FOOD Active Multilayer packaging based on Optimized PLA formulations for in minimally processed vegetables and fruits) funded by FP7-SME

1. **Bio-degredable polymers (less impact on the environment)**
2. **Transparent**
3. **Print friendly**
4. **Flexible and rigid packing as per requirement**
5. **It prevent water passage – More shelf life**
6. **It allows Breathing**

Eco-Friendly, Long shelf life packing